

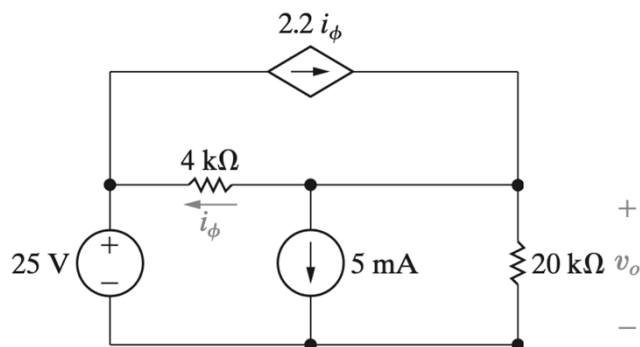
BRAC University  
Department of Electrical & Electronic Engineering  
Assignment 2, Spring 2025  
EEE/ECE101: Electrical Circuits I

**Total Marks: 120**

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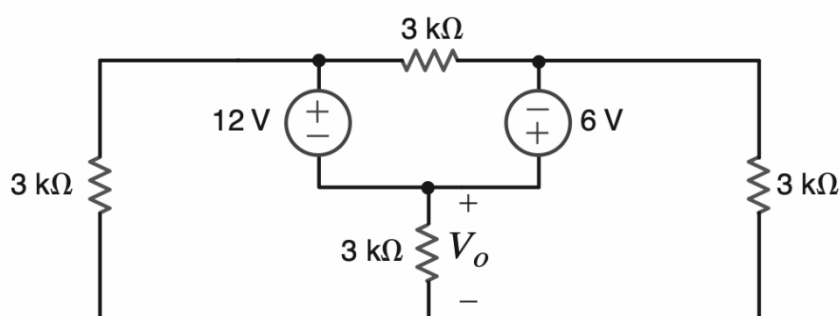
**Q1. [10 marks]**

Use the **superposition principle** to find the voltage  $v_o$  in the following circuit.



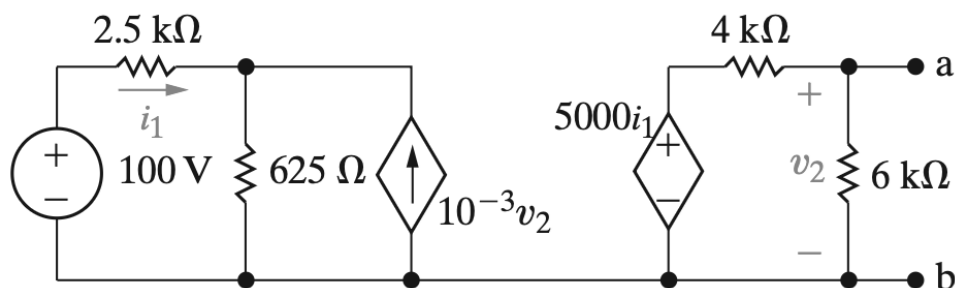
**Q2. [10 marks]**

Use the **superposition principle** to find the voltage  $V_o$  in the following circuit.



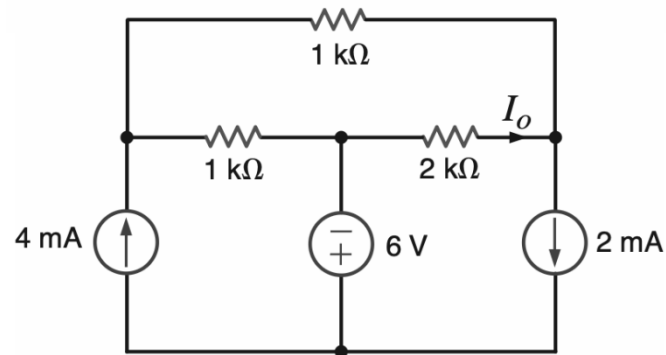
**Q3. [10 marks]**

Obtain the Thevenin equivalent circuit at terminals a-b.

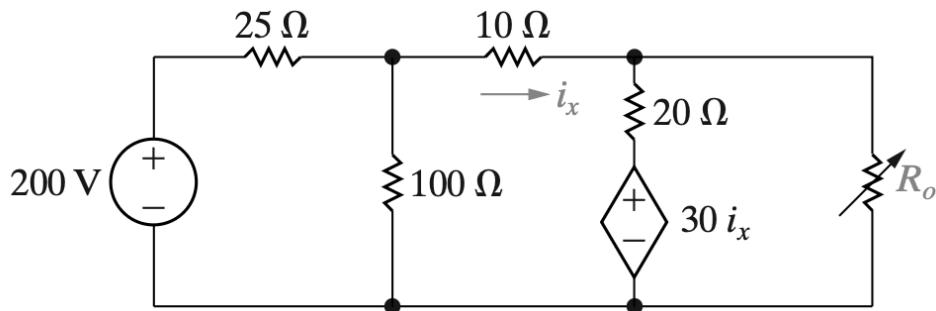


**Q4. [10 marks]**

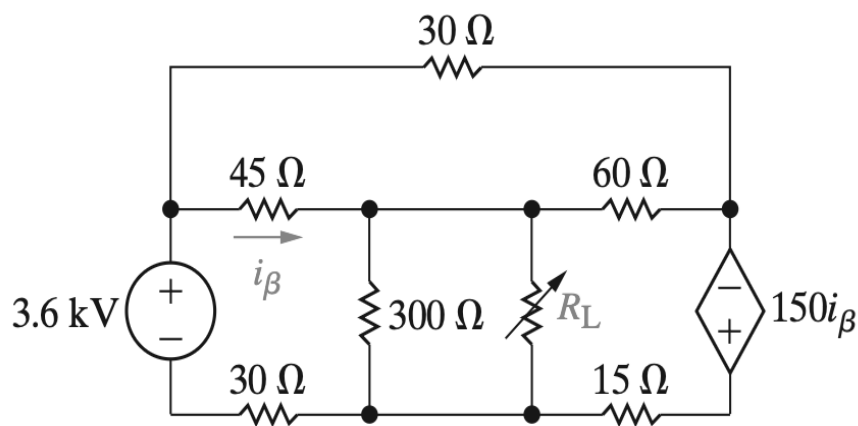
Use the **Thevenin's theorem** to find the current  $I_o$  in the following circuit.

**Q5. [10 marks]**

The variable resistor ( $R_o$ ) in the following circuit is adjusted until the power dissipated in the resistor is 250 W. Find the values of  $R_o$  that satisfy this condition.

**Q6. [10 marks]**

The variable resistor ( $R_L$ ) in the following circuit is adjusted for maximum power transfer to  $R_L$ . Find the value of  $R_L$  and the maximum power transferred to  $R_L$ .

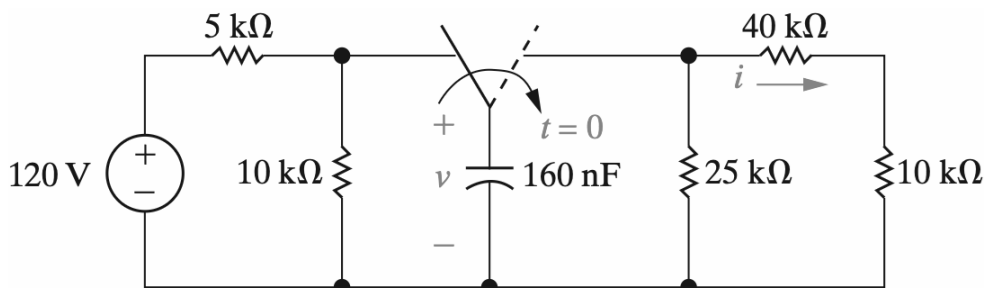


**Q7. [10 marks]**

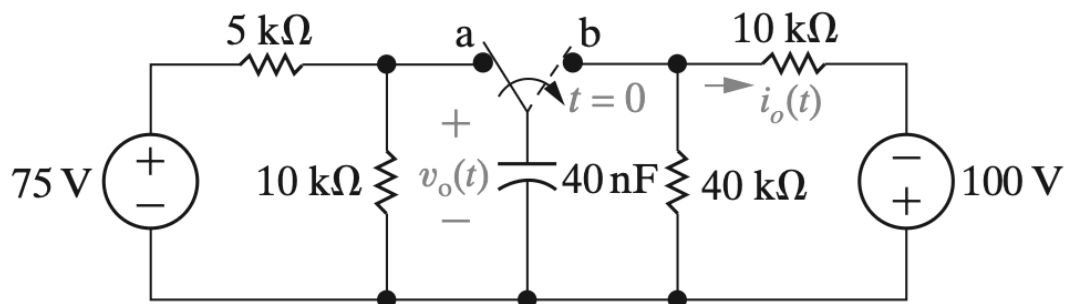
The switch in the following circuit has been in the left position for a long time. At  $t = 0$  it moves to the right position and stays there.

a) Write the expression for the capacitor voltage,  $v(t)$ , for  $t \geq 0$ .

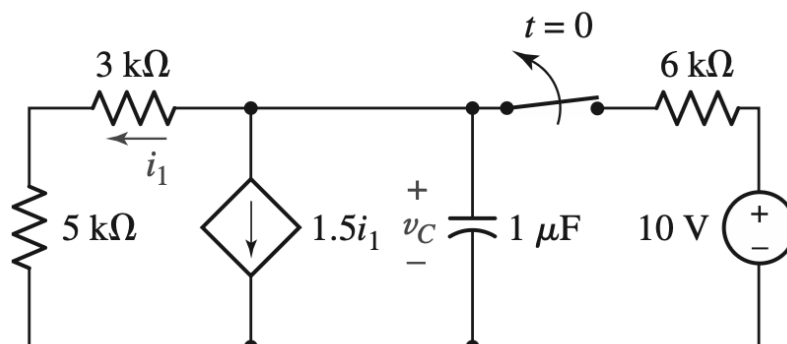
b) Write the expression for the current through the  $40 \text{ k}\Omega$  resistor,  $i(t)$ , for  $t \geq 0$ .

**Q8. [10 marks]**

The switch in the following circuit has been in position a for a long time. At  $t = 0$  the switch moves instantaneously to position b. For  $t \geq 0$ , find  $v_o(t)$  and  $i_o(t)$ .

**Q9. [10 marks]**

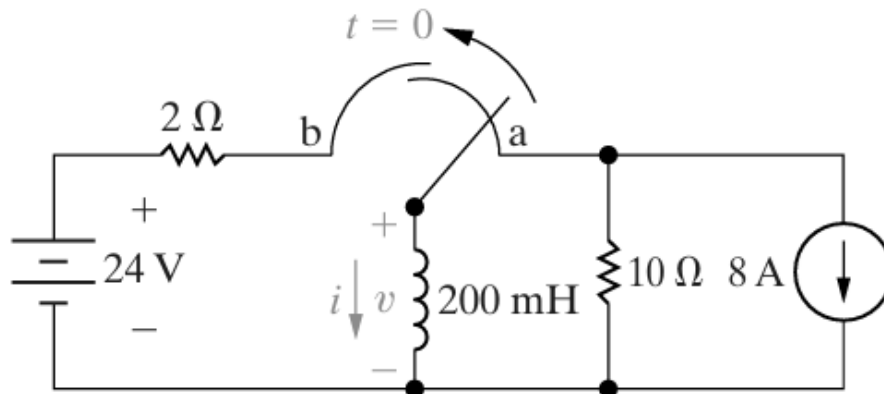
Consider the switch in the following circuit has been closed for a long time and opened at  $t = 0$ . Find  $v_c(t)$  for  $t \geq 0$ .



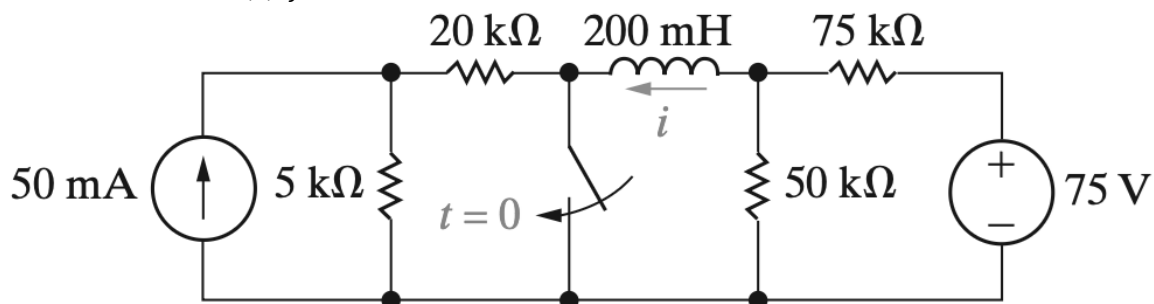
**Q10. [10 marks]**

The switch in the following circuit has been in position a for a long time. At  $t = 0$ , the switch moves from position a to position b.

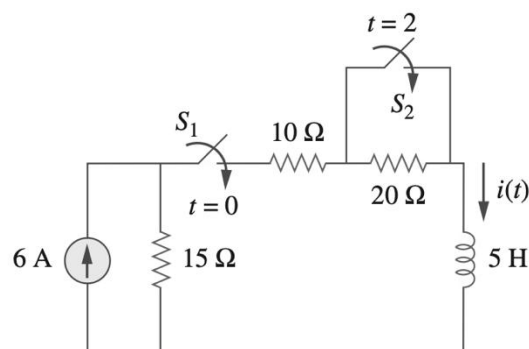
- Find the expression for  $i(t)$  for  $t \geq 0$ .
- What is the initial voltage across the inductor just after the switch has been moved to position b?
- How many milliseconds after the switch has been moved does the inductor voltage equal 24V?

**Q11. [10 marks]**

After the switch in the following circuit has been open for a long time, it is closed at  $t = 0$ . Calculate  $i(t)$  for  $t \geq 0$ .

**Q12. [10 marks]**

Switch  $S_1$  in the circuit is closed at  $t = 0$  sec and switch  $S_2$  is closed at  $t = 2$  sec. Calculate  $i(t)$  for  $t \geq 0$  and  $t < 0$ . Find  $i(1)$  and  $i(3)$ .





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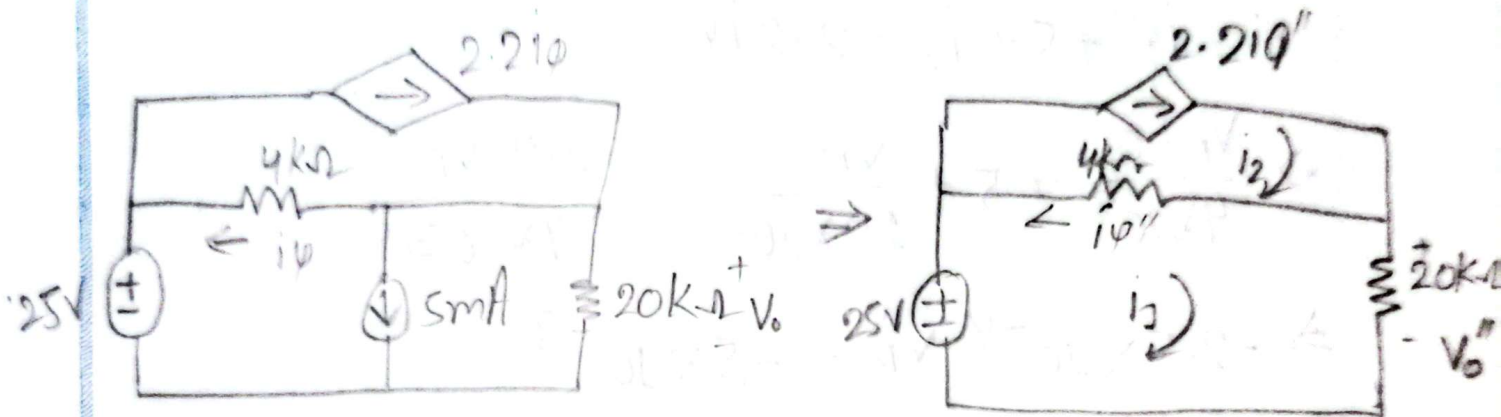
## Assignment-2

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ID: 24121205

Section: 05

# Answer to the Q. No-1



For mesh-1,

$$-25 + 4(i_1 - i_2) + 20i_1 = 0$$

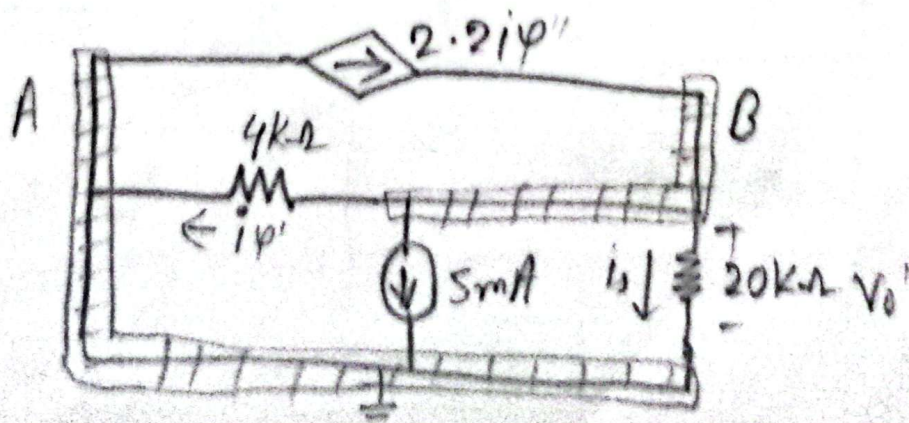
$$\Rightarrow 24i_1 - 4i_2 = 25 \dots \textcircled{i}$$

for mesh-2,  $i_2 = 2.2i_\phi'' \Rightarrow i_2 = 2.2(i_1 - i_2)$

$$\Rightarrow 2.2i_1 - 1.2i_2 = 0 \dots \textcircled{ii}$$

Solving  $\textcircled{i}$  and  $\textcircled{ii} \Rightarrow i_1 = 1.5A, i_2 = 2.75A$ .

$$\therefore V_0'' = 20 \times 1.5 = 30V$$





Applying KCL at node B  $\Rightarrow$

$$i\varphi' + 5 + i_1 = 2.2 i\varphi'$$

$$\Rightarrow \frac{V_B - 0}{4 \times 10^3} + 5 + \frac{V_B}{20 \times 10^3} = \frac{2.2 V_B}{4 \times 10^3}$$

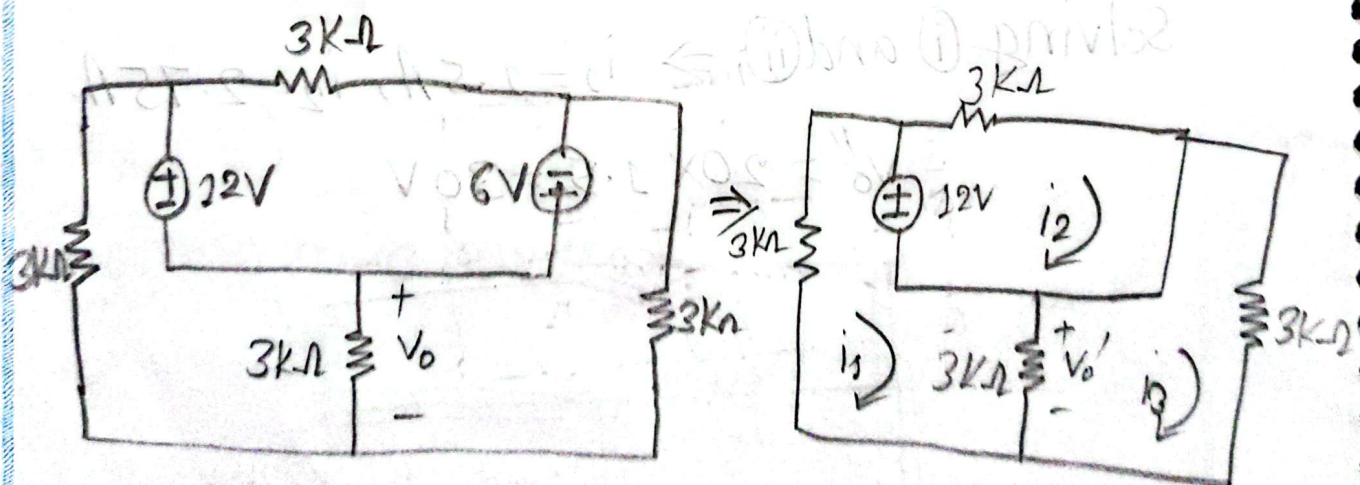
$$\Rightarrow -2.5 \times 10^{-4} V_B = -5 \times 10^{-3}$$

$$\therefore V_B = 20V$$

$$V_0' = V_B - V_A = 20 - 0 = 20V.$$

$$\therefore V_0 = V_0' + V_0'' = (30 + 20) = 50V \text{ Ans}$$

Answer to the Q. No-2



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Applying KVL at mesh-1,

$$3i_1 + 12 + 3(i_1 - i_3) = 0$$

$$\Rightarrow 6i_1 - 3i_3 = -12 \dots \textcircled{i}$$

Applying KVL at mesh 2,

$$-12 = 3i_2 \Rightarrow i_2 = -4A.$$

Applying KVL at mesh 3,

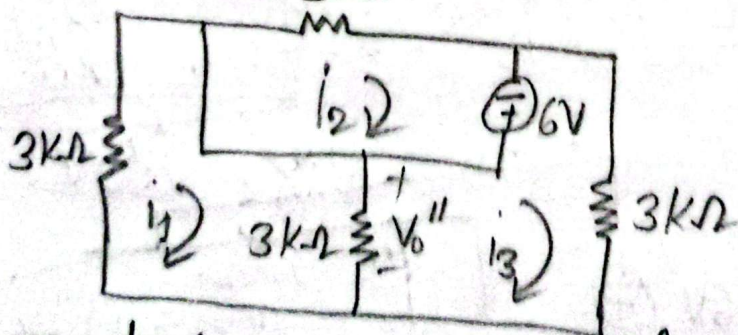
$$3i_3 + 3(i_3 - i_1) = 0$$

$$\Rightarrow 6i_3 - 3i_1 = 0 \dots \textcircled{ii}$$

Solving  $\textcircled{i}$  and  $\textcircled{ii} \Rightarrow i_1 = -\frac{8}{3}A, i_3 = -\frac{4}{3}A.$

$$\therefore V_0' = 3(i_1 - i_3) = 3\left(-\frac{8}{3} + \frac{4}{3}\right)$$

$$3k\Omega = -4V \quad \text{✓}$$



for mesh-1,

$$3i_1 + 3(i_1 - i_3) = 0$$

$$\Rightarrow 6i_1 - 3i_3 = 0 \dots \textcircled{i}$$

for mesh-2

$$3i_2 - 6 = 0$$

$$i_2 = 2A$$



for mesh-3,

$$3i_3 + 3(i_3 - i_2) + 6 = 0$$

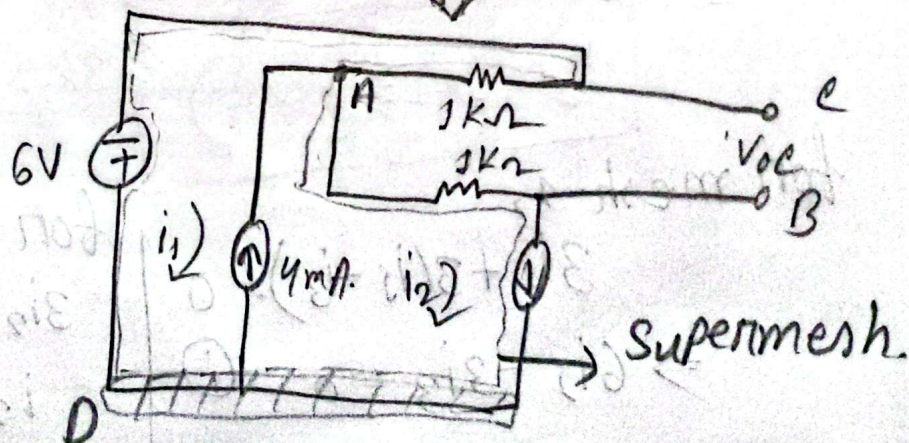
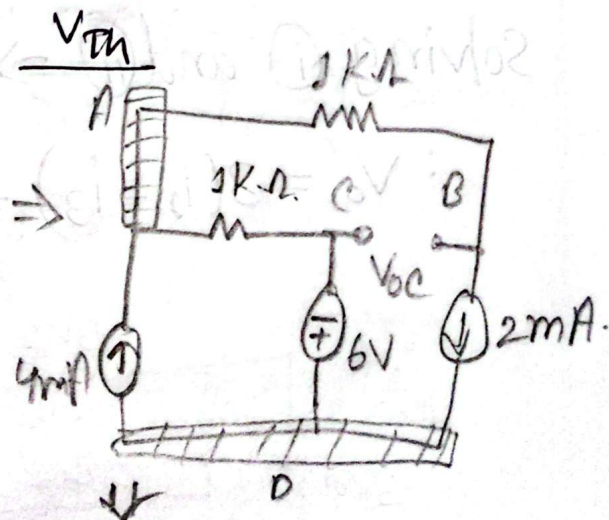
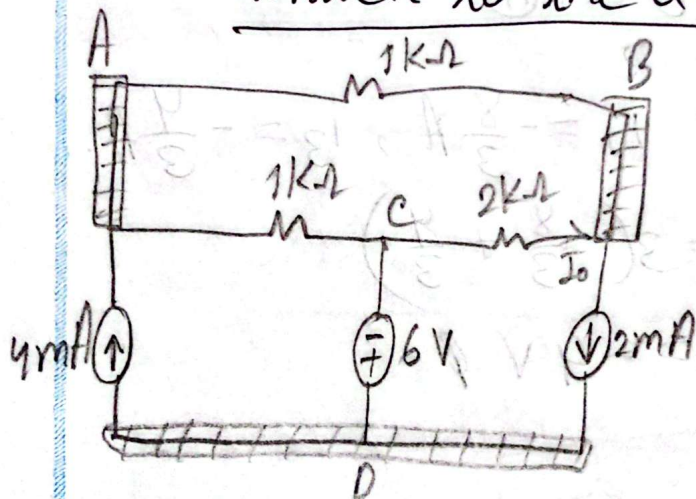
$$\Rightarrow 6i_3 - 3i_2 = -6 \quad \text{--- (ii)}$$

solving (i), (ii)  $\Rightarrow i_2 = -\frac{2}{3} A, i_3 = -\frac{4}{3} A.$

$$\therefore V_o'' = 3(i_2 - i_3) = 3\left(-\frac{2}{3} + \frac{4}{3}\right) = 2V$$

$$\therefore V_o = V_o' + V_o'' = -4 + 2 = -2V \quad \text{Ans}$$

Answer to the Q. NO-4



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From mesh-2,  $i_2 = 2 \text{ mA}$ .

From Supermesh,

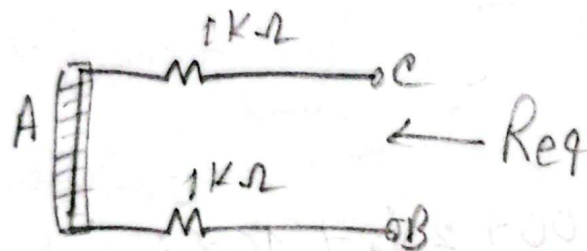
$$i_2 - i_1 = 4 \text{ mA}$$

$$\Rightarrow 2 - i_1 = 4$$

$$\therefore i_1 = -2 \text{ mA}$$

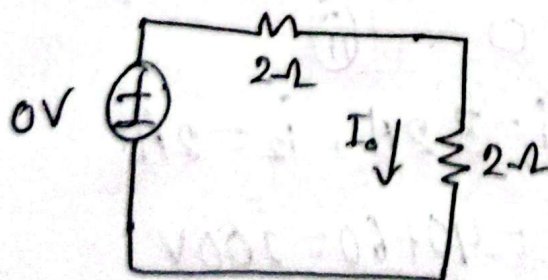
$$\therefore V_{oc} = 1(-2) + (1 \times 2) = -2 + 2 = 0 \text{ V}$$

$R_{Th}$



$$\therefore R_{eq} = 1 + 1 = 2 \text{ k}\Omega$$

Thevenin equivalent circuit  $\rightarrow$



$$\therefore I_o = \frac{0}{2+2} = 0 \text{ A}$$

Ans

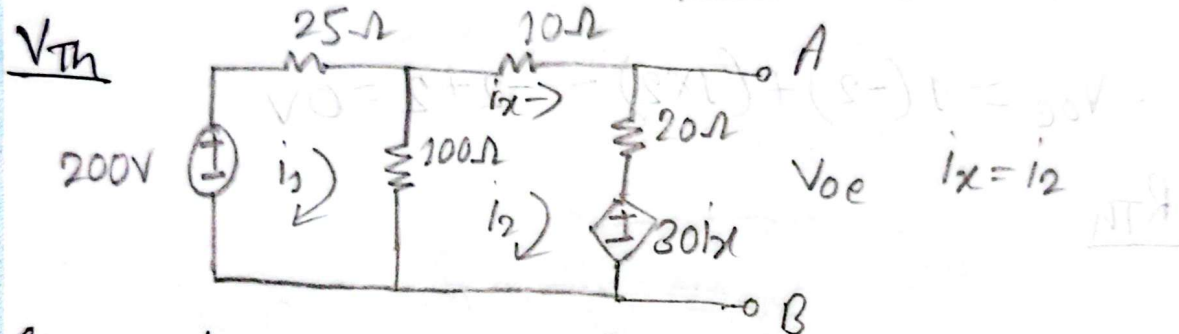
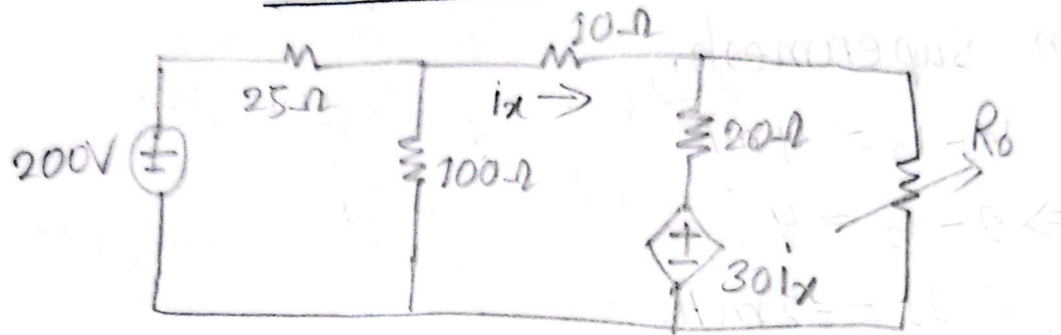


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Answer to the Q. No-5

for mesh 1  $\rightarrow -200 + 25i_1 + 10(i_1 - i_2) = 0$

$\Rightarrow 125i_1 - 100i_2 = 200 \dots \textcircled{i}$

for mesh 2  $\rightarrow 10i_2 + 20i_2 + 30i_2 + 10(i_2 - i_1) = 0$

$\Rightarrow 100i_1 - 160i_2 = 0 \dots \textcircled{ii}$

solving  $\textcircled{i}$  and  $\textcircled{ii} \Rightarrow i_1 = 3.2A, i_2 = 2A$

$\therefore V_{oc} = 20i_2 + 30i_2 = 40 + 60 = 100V$

$V_{Th} = V_{oc} = 100V$

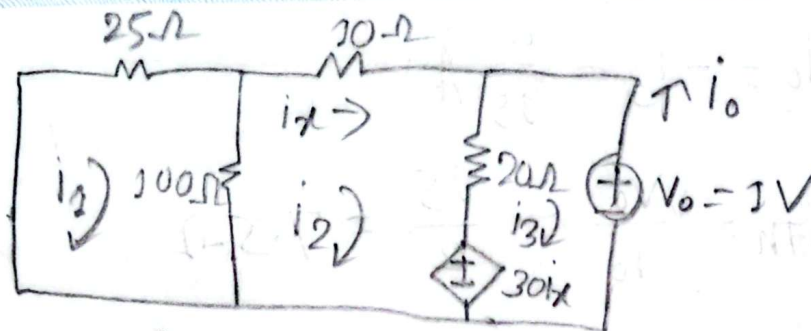
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Applying KVL at mesh-1,

$$-200 + 25i_1 + 100(i_1 - i_2) = 0$$

$$\Rightarrow 125i_1 - 100i_2 = 200 \dots \textcircled{i}$$

Applying KVL at mesh-2,

$$100(i_2 - i_1) + 10i_2 + 20(i_2 - i_3) + 30i_2 = 0$$

$$\Rightarrow -100i_1 + 160i_2 - 20i_3 = 0 \dots \textcircled{ii}$$

$$100(i_2 - i_1) + 10i_2 + 20(i_2 - i_3) + 30i_2 = 0$$

$$\Rightarrow -100i_1 + 160i_2 - 20i_3 = 0 \dots \textcircled{ii}$$

Applying KVL at mesh-3,

$$-30i_2 + 20(i_3 - i_2) + 1 = 0$$

$$\Rightarrow -50i_2 + 20i_3 = -1 \dots \textcircled{iii}$$

from  $\textcircled{i}, \textcircled{ii}, \textcircled{iii}$

$$i_1 = -\frac{2}{15} \text{ A}, \quad i_2 = -\frac{1}{30} \text{ A}, \quad i_3 = -\frac{2}{15} \text{ A}$$



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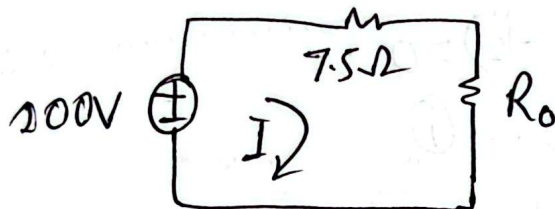
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$$\therefore i_0 = -i_3 = \frac{2}{15} A$$

$$R_{Th} = \frac{V_0}{i_0} = \frac{15}{2} = 7.5 \Omega$$

Thevenin's equivalent circuit  $\Rightarrow$



$$P_0 = I^2 R_0 \Rightarrow P_0 = \left( \frac{V_{Th}}{R_{Th} + R_0} \right)^2 R_0$$

$$\Rightarrow 250 = \left( \frac{100}{7.5 + R_0} \right)^2 R_0$$

$$\Rightarrow 250 = \frac{10000 R_0}{56.25 + 15 R_0 + R_0^2}$$

$$\Rightarrow 14125 + 3750 R_0 + 250 R_0^2 = 10000 R_0$$

$$\Rightarrow 250 R_0^2 - 6250 R_0 + 14125 = 0$$

$$\Rightarrow R_0 = \frac{-(-6250) \pm \sqrt{(-6250)^2 - 4 \cdot 250 \cdot 14125}}{2 \times 250}$$

$$\Rightarrow R_0 = \frac{6250 \pm 4993.746}{500}$$

$$\therefore R_0 = 22.5 \Omega, 2.5 \Omega$$

Ans

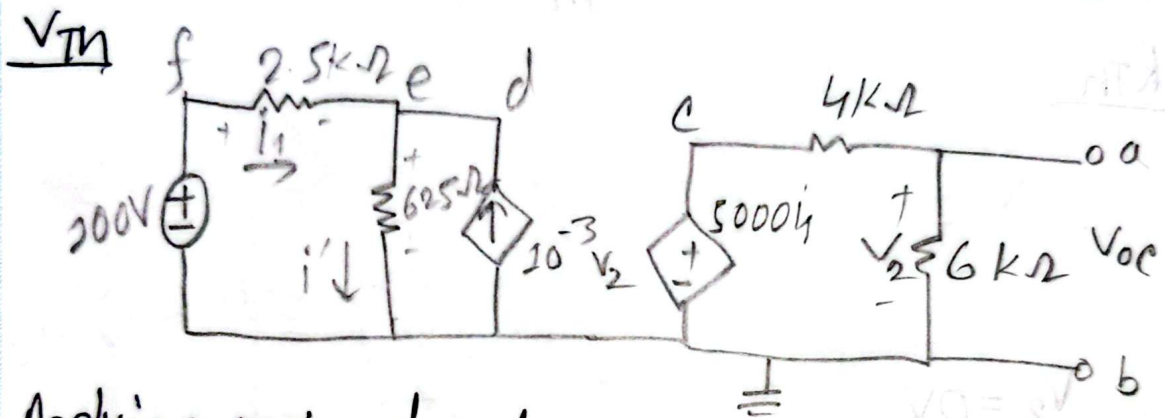
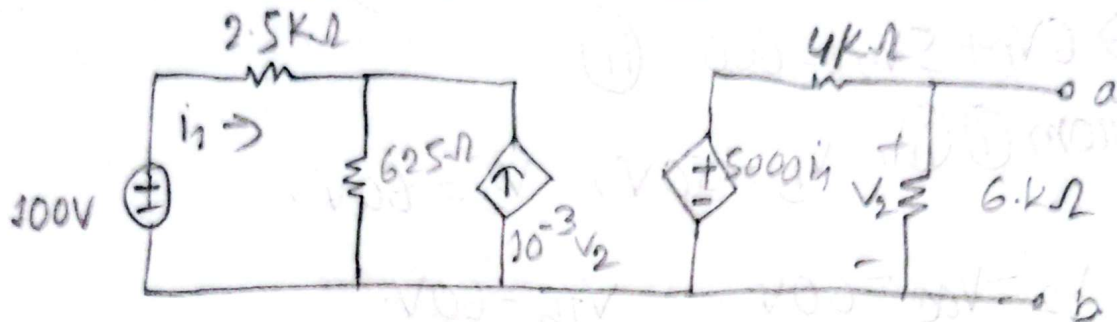
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### Ans to the Q. No-3



Applying KVL at node e,

$$i_1 + 10^{-3} V_2 = i'$$

$$\Rightarrow \frac{V_A - V_B}{2500} + 10^{-3} V_2 = \frac{V_B}{625}$$

$$\Rightarrow \frac{V_A - V_B + 2.5 V_2}{2500} = \frac{V_B}{625}$$

$$\Rightarrow (100 - V_B + 2.5 V_2) \times 625 = 2500 V_B$$

$$\Rightarrow -3125 V_B + 1562.5 V_2 = -62500 \dots \textcircled{1}$$

$$\Rightarrow V_2 = \frac{6}{10} \times 5000 i_1$$



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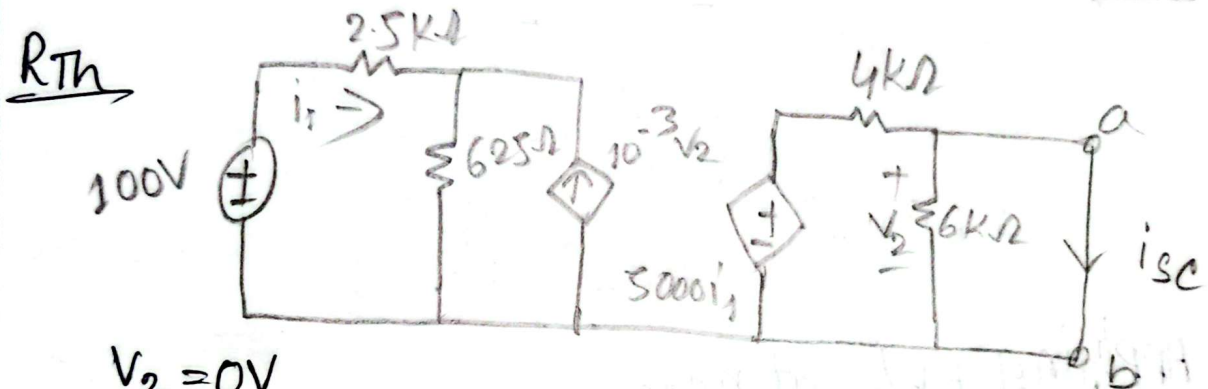
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$$\Rightarrow 10V_2 = 30000i_1 - 30000\left(\frac{V_A - V_B}{2500}\right)$$

$$\Rightarrow 6V_B + 5V_2 = 600 \dots (ii)$$

from (i) (ii),  $V_B = 50V$ ,  $V_2 = 60V$ ,

$$\therefore V_2 = V_{oc} = 60V, \quad V_{Th} = 60V.$$

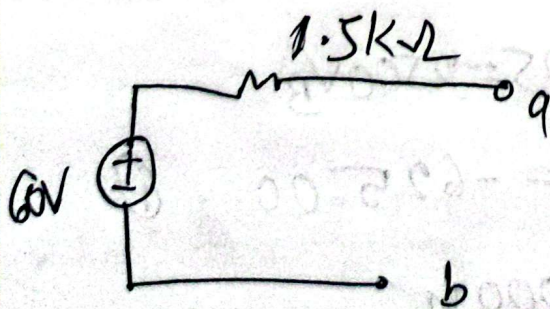


$$V_2 = 0V$$

$$\therefore i_1 = \frac{100}{2500 + 625} = 0.032A.$$

$$i_{sc} = \frac{5000 i_1}{4000} = \frac{5000 \times 0.032}{4000} = 0.04$$

$$R_{Th} = \frac{V_{Th}}{i_{sc}} = \frac{60}{0.04} = 1500\Omega = 1.5K\Omega$$



Thevenin  
equivalent  
circuit

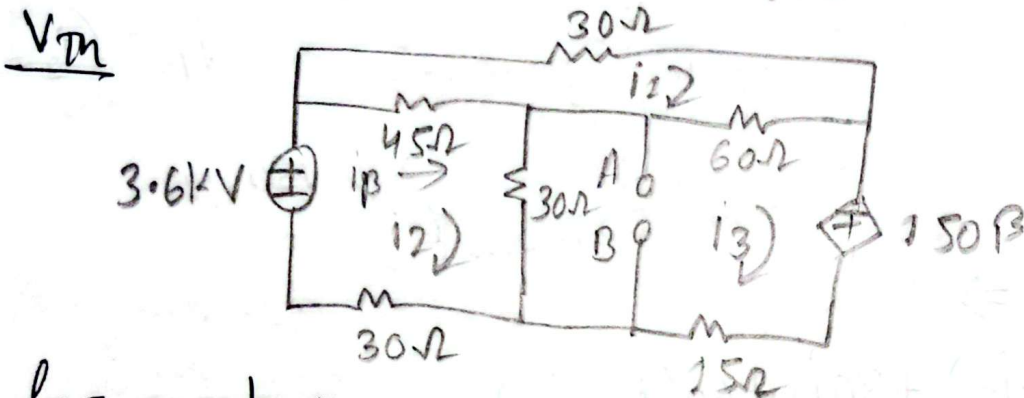
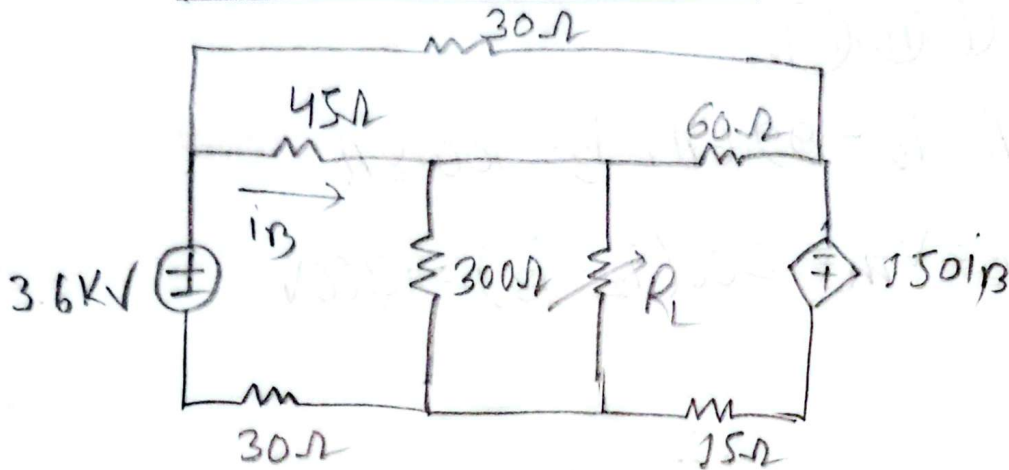


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Ans to the Q. NO-6

for mesh-1,

$$30i_1 + 60(i_1 - i_3) + 45(i_1 - i_2) = 0$$

$$\Rightarrow 135i_1 - 45i_2 - 60i_3 = 0 \quad \text{--- (i)}$$

for mesh-2,

$$45(i_2 - i_1) + 300(i_2 - i_3) + 30i_2 - 3600 = 0$$

$$\Rightarrow -45i_1 + 375i_2 - 300i_3 = 3600 \quad \text{--- (ii)}$$

for mesh-3,

$$-150(i_2 - i_1) + 15i_3 + 300(i_3 - i_2) + 60(i_3 - i_1) = 0$$

$$\Rightarrow -150i_2 + 150i_1 + 15i_3 + 300i_3 - 300i_2 + 60i_3 - 60i_1 = 0$$

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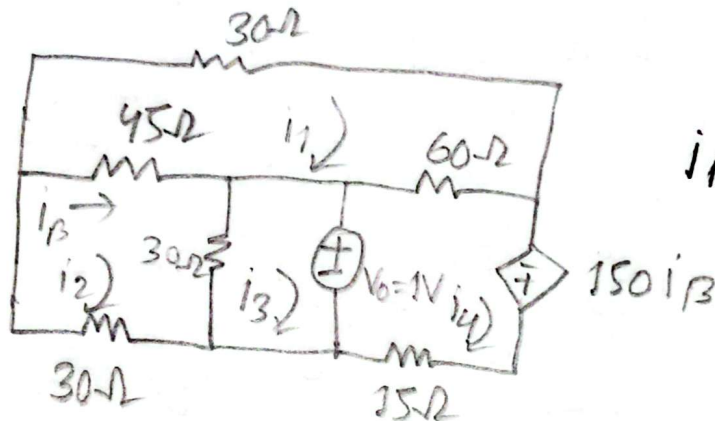
$$\Rightarrow 90i_1 - 450i_2 + 375i_3 = 0 \quad \text{--- (ii)}$$

from (i), (ii), (iii),

$$i_1 = 78A, \quad i_2 = 99.6A, \quad i_3 = 100.8A.$$

$$V_{oc} = V_{th} = 300(i_2 - i_3) = -360V$$

R<sub>th</sub>



$$i_B = (i_2 - i_3)$$

mesh-1  $\Rightarrow$

$$45(i_1 - i_2) + 30i_1 + 60(i_1 - i_4) = 0$$

$$\text{mesh} \Rightarrow 135i_1 - 45i_2 - 60i_4 = 0 \quad \text{--- (i)}$$

mesh-2,

$$30i_2 + 45(i_2 - i_1) + 300(i_2 - i_3) = 0$$

$$\Rightarrow -45i_1 + 375i_2 - 300i_3 = 0 \quad \text{--- (ii)}$$

mesh-3,

$$300(i_3 - i_2) + 1 = 0$$

$$\Rightarrow 300i_2 - 300i_3 = 1 \quad \text{--- (iii)}$$



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mesh-4,

$$15i_4 - 1 + 60(i_4 - i_3) - 150(i_2 - i_3) = 0$$

$$\Rightarrow 15i_4 - 1 + 60i_4 - 60i_3 - 150i_2 + 150i_3 = 0$$

$$\Rightarrow 90i_3 - 150i_2 + 75i_4 = 1 \dots \textcircled{IV}$$

$$\textcircled{IV} \Rightarrow i_4 = \frac{1 - 90i_3 + 150i_2}{75}$$

$$\textcircled{I} \Rightarrow 135i_3 - 45i_2 - \frac{4(1 - 90i_3 + 150i_2)}{5} = 0$$

$$\Rightarrow 675i_3 - 225i_2 - 4 + 360i_3 - 600i_2 = 0$$

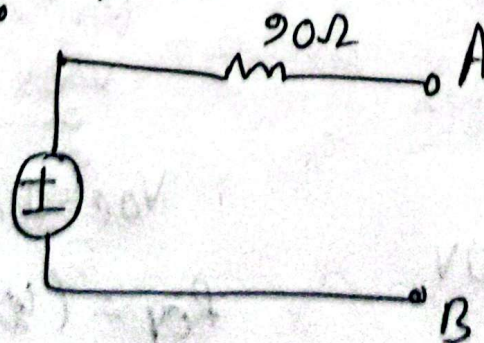
$$\Rightarrow 1035i_3 - 825i_2 = 4 \dots \textcircled{V}$$

from,  $\textcircled{I}, \textcircled{II}, \textcircled{V}$ ,  $i_1 = -\frac{7}{540} \text{ A}$ ,  $i_2 = -\frac{19}{900} \text{ A}$ ,  $i_3 = -\frac{11}{450} \text{ A}$

$$i_4 = -\frac{1}{7} \text{ A}$$

$$\therefore i_o = i_4 - i_3 = \frac{1}{90} \text{ A}$$

$$R_{Th} = \frac{V_o}{I_o} = 90 \Omega$$



Thevenin equivalent circuit.



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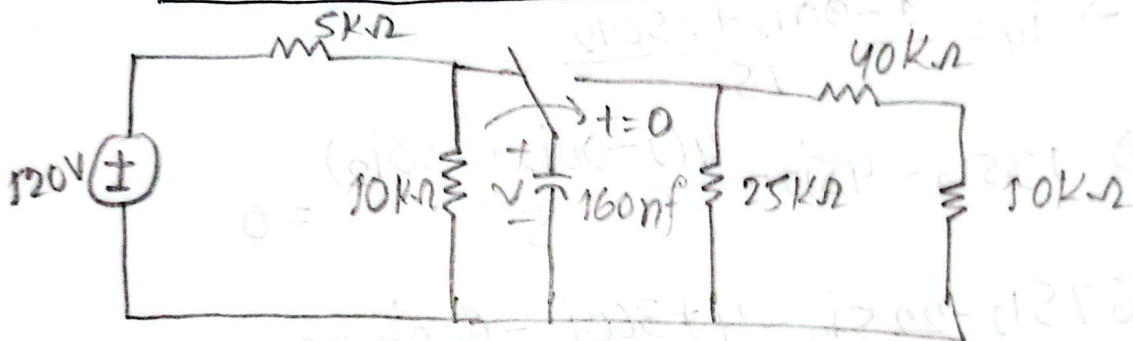
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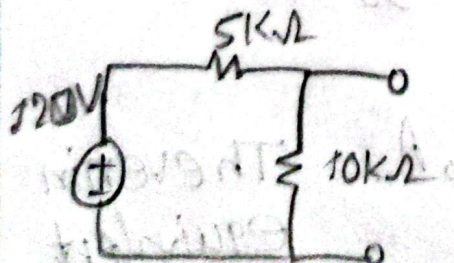
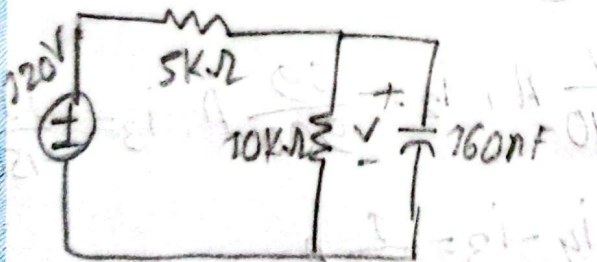
∴ The ~~power~~ maximum power transferred to  $R_L$

$$P = \frac{V_{Th}^2}{4 \cdot R_{Th}} = \frac{(-360)^2}{4 \times 90} = 360 \text{ W. Ans}$$

Answer to the Q. NO-7



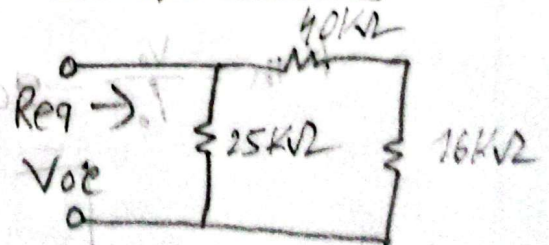
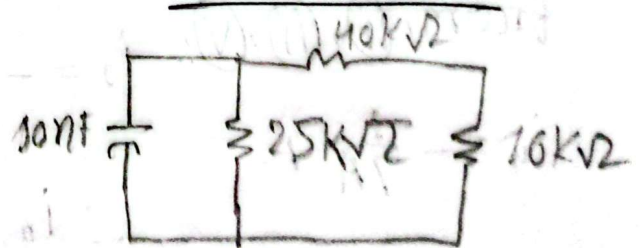
Before switching



$$V_{oe} = \frac{10}{5+10} \times 120 = 80 \text{ V}$$

$$V_{oe} = (V_o^-) = 80 \text{ V}$$

After switching



$$V_{oe} = 0, \quad V(\alpha) = 0 \text{ V.}$$

$$R_{eq} = (40+20) \parallel 25 = \frac{50}{3} \text{ k}\Omega.$$

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$$\tau = R_{eq} \cdot C = \frac{50}{3} \times 10^3 \times 160 \times 10^{-9} = \frac{1}{375}$$

Now,

$$v(t) = v(\infty) + [v(0) - v(\infty)] e^{-\frac{t}{\tau}}$$

$$\Rightarrow v(t) = 0 + [80 - 0] e^{-\frac{t \times 375}{1}}$$

$$\Rightarrow v(t) = 80 e^{-375t} \quad \underline{\text{Ans}}$$

$$i(t) = C \frac{dv}{dt} = C \frac{d}{dt} (80 e^{-375t})$$

$$i(t) = 160 \times 10^{-9} \times 80 \times (-375) e^{-375t}$$

$$i(t) = -4.8 \times 10^{-3} e^{-375t}$$

The current through the  $40k\Omega$  resistor:

$$i(t) = \frac{-25}{25+50} \times (-4.8 \times 10^{-3} e^{-375t})$$

$$i(t) = 1.6 \times 10^{-3} e^{-375t} \text{ A.}$$

$$i(t) = 1.6 e^{-375t} \text{ mA} \quad \underline{\text{Ans}}$$



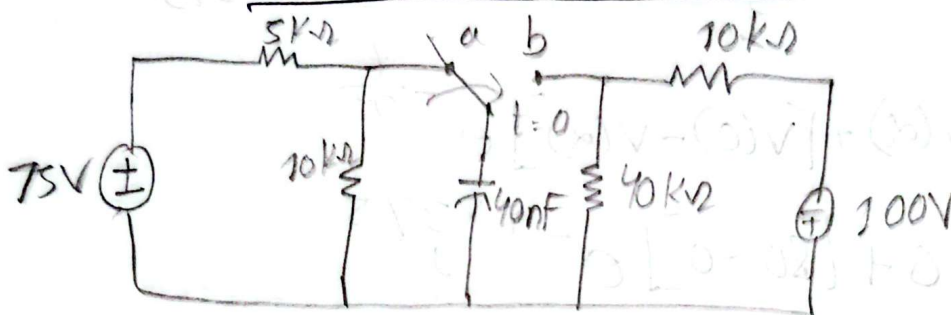
TOPIC NAME :

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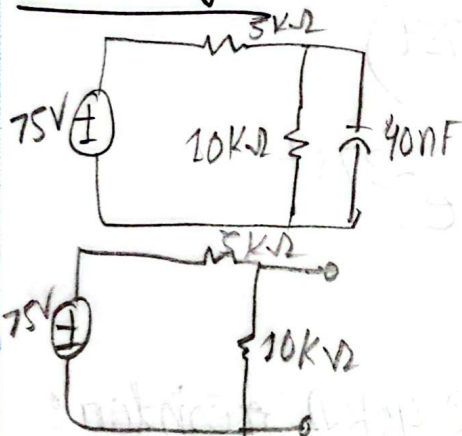
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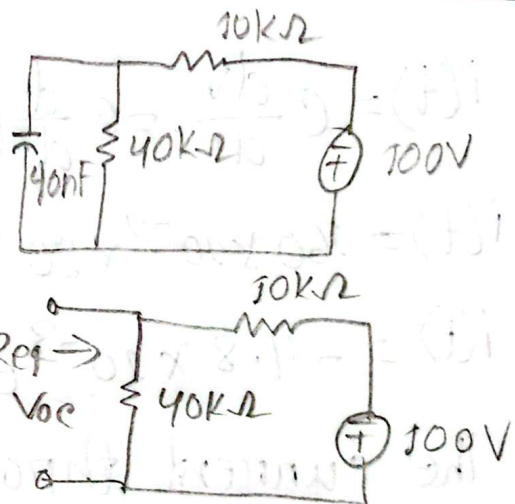
# Answer to the Q. NO-8



Before switching



After switching



$$V_{oc} = \frac{10}{5+10} \times 75$$

$$= 50V$$

$$V_o(0) = 50V$$

$$R_{eq} = 40 \parallel 10 = \frac{40 \times 10}{40+10} = 8k\Omega$$

$$V_{oc} = \frac{40}{40+10} \times (-100) = -80V$$

$$V_o(\infty) = -80V$$

$$\tau = R_{eq} \cdot C$$

$$= 8 \times 10^3 \times 40 \times 10^{-9}$$

$$= \frac{1}{3125} s$$

TOPIC NAME :

DAY :

TIME :

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$$V_o(t) = V_o(\infty) + [V_o(0) - V_o(\infty)] e^{-\frac{t}{\tau}}$$

$$V_o(t) = -80 + [50 + 80] e^{-\frac{t \times 3125}{1}}$$

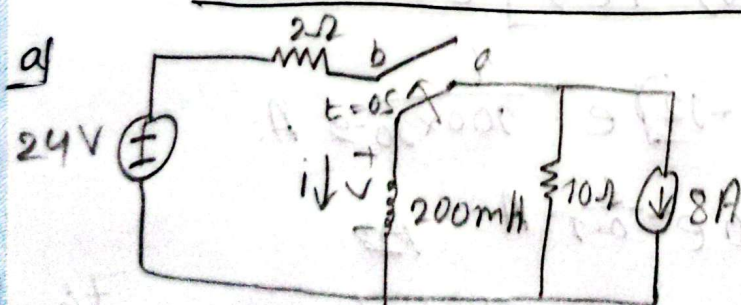
$$V_o(t) = -80 + 130 e^{-3125t} \quad \text{Ans}$$

$$i_o(t) = \frac{V_o(t) - (-100)}{10} = \frac{(-80 + 130 e^{-3125t}) + 100}{10}$$

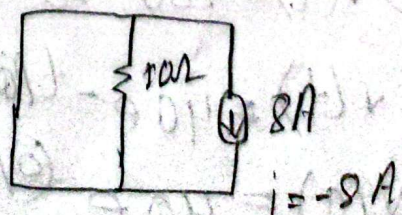
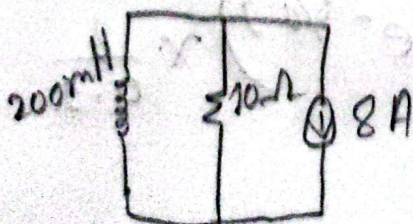
$$\Rightarrow i_o(t)' = \frac{20 + 130 e^{-3125t}}{10}$$

$$i_o(t) = 130 e^{-3125t} + 2 \text{ mA} \quad \text{Ans}$$

Ans to the Q. No-10



$t < 0$





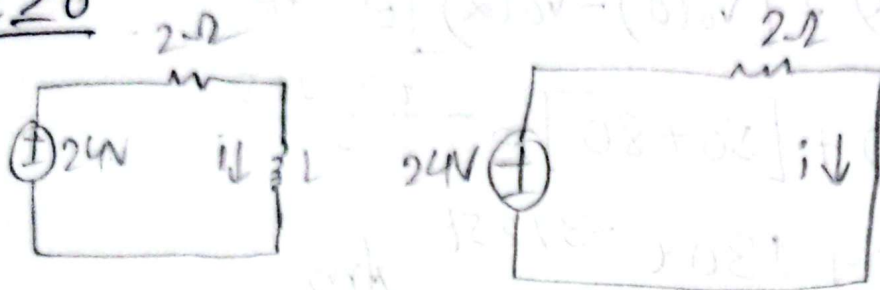
TOPIC NAME :

DAY

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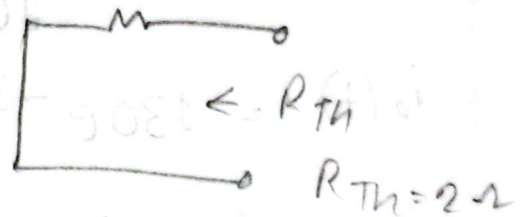
$t \geq 0$



$$i(\infty) = \frac{24}{2} = 12 \text{ A}$$

$$i(t_0) = i(t_0^-) = -8 \text{ A}$$

$$\tau = \frac{L}{R_{Th}} = \frac{200 \times 10^{-3}}{2} = 100 \times 10^{-3}$$



$$i(t) = i(\infty) + [i(t_0) - i(\infty)] e^{-\frac{t}{\tau}}$$

$$\Rightarrow i(t) = 12 + [-8 - 12] e^{-\frac{t}{100 \times 10^{-3}}} \text{ A}$$

$$i(t) = 12 - 20 e^{-\frac{t}{0.1}} \text{ A} \quad \underline{\text{Ans}}$$

$$b) v(t) = \frac{di}{dt} = 200 \times 10^{-3} \times \frac{d}{dt} (12 - 20 e^{-\frac{t}{0.1}})$$

$$\Rightarrow v(t) = 200 \times 10^{-3} \times (20 e^{-\frac{t}{0.1}}) \times \frac{1}{0.1}$$

$$v(t) = 40 e^{-\frac{t}{0.1}} \text{ V}$$

$$v(0) = 40 e^{-(0/0.1)} = 40 \text{ V} \quad \underline{\text{Ans}}$$

TOPIC NAME : \_\_\_\_\_

DAY : \_\_\_\_\_

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$$c) 24 = 40e^{-10t}$$

$$\Rightarrow e^{-10t} = \frac{24}{40}$$

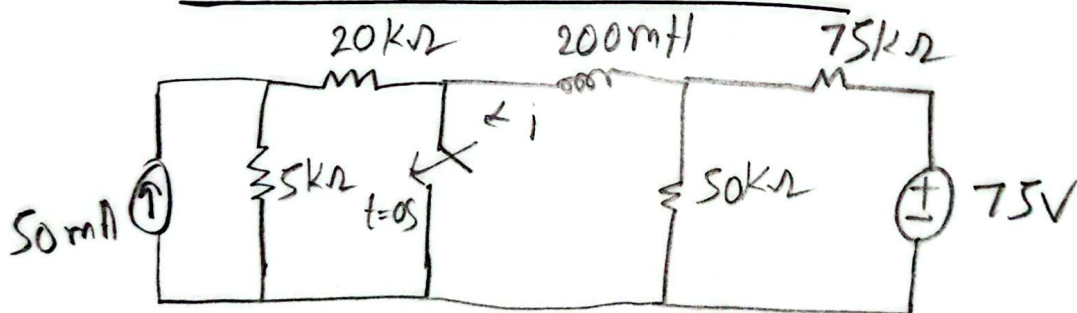
$$\Rightarrow \ln e^{-10t} = \ln \frac{24}{40}$$

$$\Rightarrow -10t = \ln\left(\frac{24}{40}\right)$$

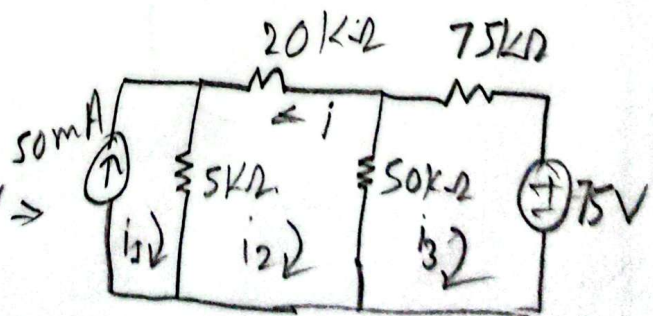
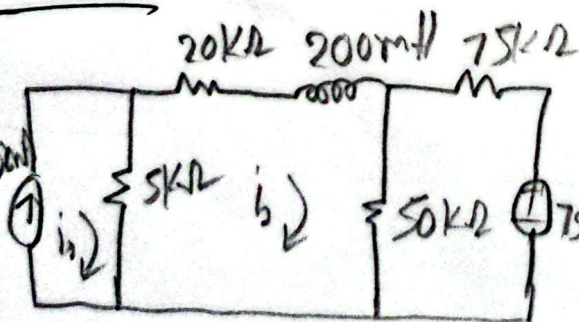
$$\Rightarrow t = \frac{\ln(24/40)}{-10}$$

$$t = 0.05108 \text{ s Ans}$$

Ans to the Q. No-11



$t < 0$





$$\text{mesh-1} \Rightarrow i_1 = 50 \text{ mA}$$

$$\text{mesh-2} \Rightarrow 5(i_2 - i_1) + 20i_2 + 50(i_2 - i_3) = 0$$

$$\Rightarrow 75i_2 - 50i_3 = 5i_1$$

$$\Rightarrow 75i_2 - 50i_3 = 0.250 \quad \text{--- (i)}$$

$$\text{mesh-3} \Rightarrow$$

$$50(i_3 - i_2) + 75i_3 + 75 = 0$$

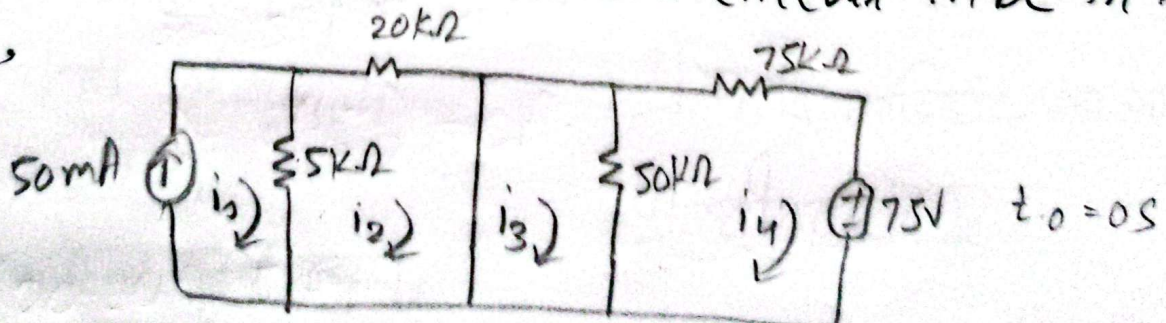
$$\Rightarrow -50i_2 + 125i_3 = -75 \quad \text{--- (ii)}$$

from (i), (ii),  $i_2 = 4 \text{ mA}$ ,  $i_3 = 1 \text{ mA}$ .

$$\therefore i = -i_2 = -4 \text{ mA}$$

$t > 0$

The inductor acts as a short circuit in DC in steady state.



$$\text{mesh-1, } i_1 = 50 \text{ mA}$$

$$\text{mesh-2, } 5(i_2 - i_1) + 20i_2 = 0$$

$$\Rightarrow 25i_2 = 250 \quad \therefore i_2 = 10 \text{ mA}$$



$$\text{mesh-3} \Rightarrow 50(i_3 - i_4) = 0$$

$$\Rightarrow 50i_3 - 50i_4 = 0 \dots \textcircled{i}$$

$$\text{mesh-4} \Rightarrow 50(i_4 - i_3) + 75i_4 + 75 = 0$$

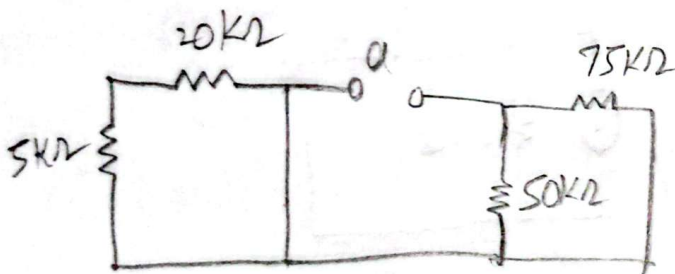
$$\Rightarrow -50i_3 + 125i_4 = -75 \dots \textcircled{ii}$$

from  $\textcircled{i}, \textcircled{ii}$ ,  $i_3 = -1 \text{ mA}$ ,  $i_4 = -1 \text{ mA}$ .

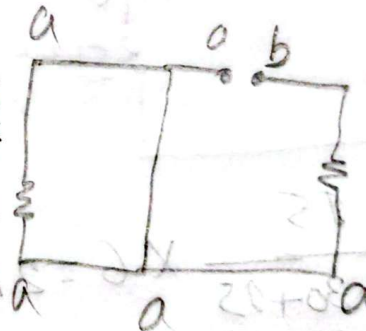
$$\therefore i = -i_3 = 1 \text{ mA}$$

$$\therefore i(\infty) = 1 \text{ mA}, i(t_0) = i(t_0^-) = i(t_0^+)$$

$$i(t_0) = -4 \text{ mA}$$



$$\Rightarrow \frac{20 \times 5}{20 + 5} = 4 \text{ k}\Omega$$



$$\frac{75 \times 50}{75 + 50} = 30 \text{ k}\Omega$$

$$\tau = \frac{L}{R_{eq}} = \frac{200 \times 10^{-3}}{30000} = \frac{1}{150000} \text{ s} \quad R_{eq} = 30 \text{ k}\Omega$$

$$i(t) = i(\infty) + [i(0) - i(\infty)] e^{-\frac{t}{\tau}}$$

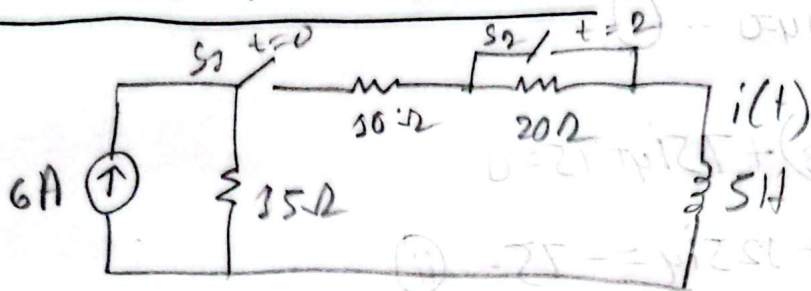
$$i(t) = 1 + [-4 - 1] e^{-\frac{t}{\tau}}$$

$$i(t) = 1 - 5e^{-150000t} \text{ A}$$

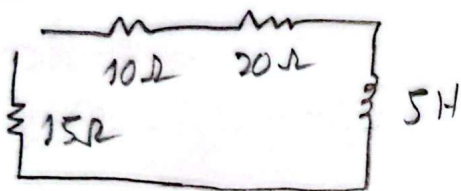
Ans  $\frac{1}{\tau} = \frac{1}{2 \times 10^{-5}} = \frac{1}{2 \times 10^{-5}} = 5 \times 10^4$



Ans to the Q.No-12

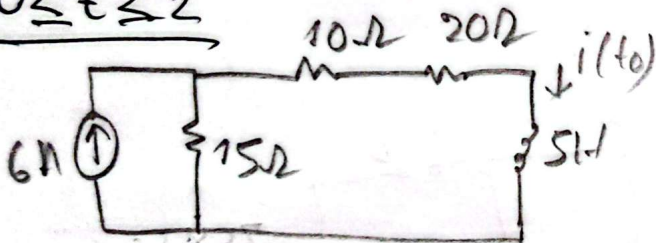


$t < 0$



$$i(t_0^-) = 0 \text{ A}$$

$0 \leq t \leq 2$

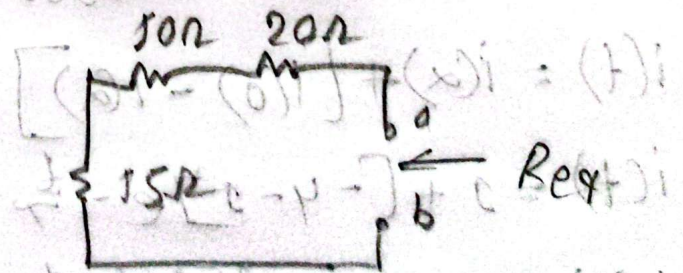
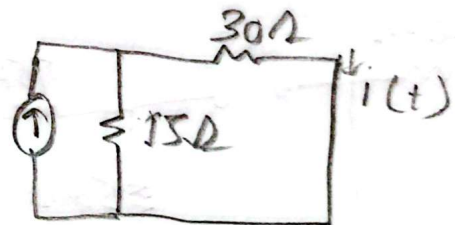


$$i(\infty) = \frac{15}{30+15} \times 6 = 2 \text{ A}$$

$$i(t_0) = i(t_0^-) = 0 \text{ A}$$

$$R_{eq} = (20+10+15) \Omega = 45 \Omega$$

$$T = \frac{L}{R_{eq}} = \frac{5}{45} = \frac{1}{9} \text{ s}$$



$$i(t) = i(\alpha) + [i(t_0) - i(\alpha)] e^{-\frac{t-t_0}{\tau}}$$

$$\Rightarrow i(t) = 2 + [0 - 2] e^{-\frac{t-0}{1/9}}$$

$$\Rightarrow i(t) = 2(1 - e^{-9t}) \text{ A.}$$

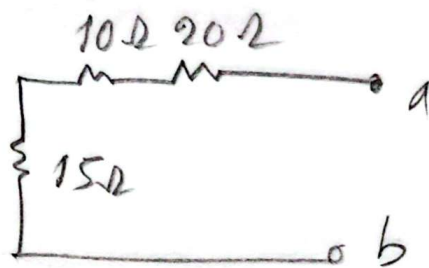
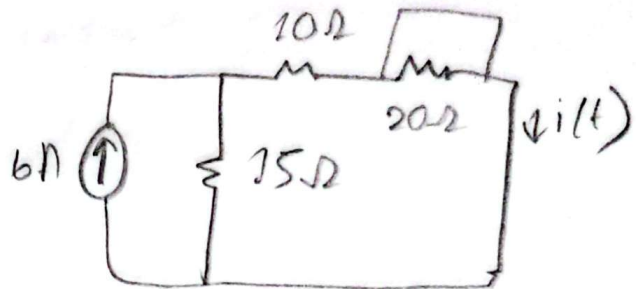
$$t > 2$$

$$i(\alpha) = \frac{15}{10+15} \times 5$$

$$= 3.6 \text{ A}$$

$$i(2) = 2 \times (1 - e^{-9 \times 2}) \approx 2 \text{ A.}$$

$$\tau = \frac{L}{R_{eq}} = \frac{5}{25} = \frac{1}{5} \text{ s.}$$



$$i(t) = i(\alpha) + [i(2) - i(\alpha)] e^{-(t-2)/\tau}$$

$$\Rightarrow i(t) = 3.6 + [2 - 3.6] e^{-5(t-2)}$$

$$\Rightarrow i(t) = 3.6 - 1.6 e^{-5(t-2)} \text{ A.}$$

$$\therefore i(t) = \begin{cases} 0 \text{ A} & , t < 0 \\ 2(1 - e^{-9t}) \text{ A} & , 0 \leq t \leq 2 \\ 3.6 - 1.6 e^{-5(t-2)} \text{ A} & , t \geq 2 \end{cases}$$

$$i(1) = 2(1 - e^{-9 \times 1}) = 1.9997 \text{ A}$$

$$i(3) = 3.6 - 1.6 e^{-5(3-2)} = 3.589 \text{ A}$$